1) 7/4/2026

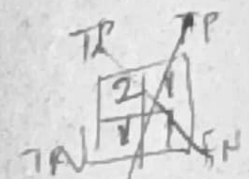
2

S	Featurel Age	F2 Income	F3 score	Actual class	predicted probability	Predict based on 0.5
S1	25	40	050	1	0.90	1 - TP
S2	30	60	700	1	0.75	1 - TP
S3	45	35	600	0	0.60	1 - FP
S4	50	30	580	0	0.40	0 - TN
S5	28	55	720	1	0.30	0 - FN

Positive → 1

Negative → 0

If $\geq (0.5) \rightarrow 1$
$< 0.5 \rightarrow 0$ } Threshold condition



Confusion matrix

predicted \ Actual	1	0
1	TP_2	FP_1
0	FN_1	TN_1

1) Accuracy

$$\frac{TP+TN}{Total} = \frac{3}{5}$$

2) precsion

$$\frac{TP}{TP+FP} = \frac{2}{3}$$

3). Recall (True positive Rate)

$$\frac{TP}{TP+FN} = \frac{2}{3} = 0.67$$

4) $\frac{FP}{FP+TN}$ = False positive rate

$$= \frac{1}{2} = 0.5$$

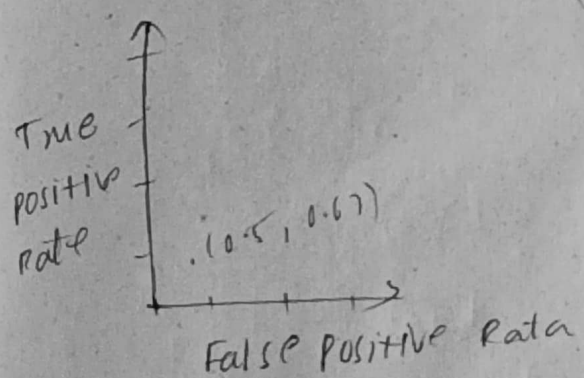
4) F1-score =

$$\frac{2 \times R \times P}{R+P}$$

$$= \frac{2 \times 2/3 \times 2/3}{\frac{2}{3} + 2/3}$$

$$= \frac{8/9}{\frac{4}{3}} = 0.\overline{6}$$

5) ROC (Receive operating characteristics)



If ihd $\geq 0.3 \rightarrow 1$
$< 0.3 \rightarrow 0$

Ac	P
1	1
1	1
1	1
0	1
0	1
1	1

	TP	FP
1	3	0
0	2	0
	FN 1	TN 0

TPR $= \frac{3}{5}$

$FP = \frac{0}{2} = 0$